DHS/MIS malaria prevalence and child mortality analysis

Main results

- **Higher malaria prevalence is associated with higher post-neonatal mortality, and this holds regardless of covariate adjustment.** Each 10 percentage-point increase in PfPR2-10 is associated with about a 8.6% increase in post-neonatal mortality (ridge-linear summary; 95% CI 5.7%–11.7%; `p<0.001`). The estimate is essentially unchanged whether we use the regional covariate block alone or the full regional-plus-national block that now includes national vaccine coverage and child HIV prevalence.
- **Neonatal mortality, the negative control, shows no such association.** The ridge-linear neonatal estimate is +1.0% per 10 PfPR points (95% CI −1.2% to +3.2%; `p=0.37`), and the selected prevalence term for neonatal mortality is effectively flat. This, together with the stability under HIV and vaccine adjustment, reduces the concern that the main result simply reflects broad differences in child survival between higher- and lower-prevalence settings.
- **A time-stable nonlinear prevalence-mortality relationship fits best.** The additive spline `s(PfPR)+s(year)` is selected by AIC over a spline time-interaction (ΔAIC 4.9; interaction `p=0.16`), a linear form (ΔAIC 10.2), and a full `te(PfPR, year)` surface (ΔAIC 11.2). It implies population-average malaria-attributable fractions of post-neonatal mortality of 14%, 34% and 40% at 10%, 30% and 50% PfPR2-10 (versus a 1% counterfactual).
- **The preferred model estimates about 482,000 malaria-attributable post-neonatal deaths for the latest period — close to IHME.** The model-parameter 95% interval is 348,000–606,000. The point estimate is about 3% higher than the IHME/GBD 2025 estimate and about 11% higher than an approximate WHO under-five figure; the intervals overlap. This is not a validation against a common estimand: the model captures direct and indirect malaria-associated post-neonatal mortality, whereas IHME and WHO assign deaths specifically to malaria.

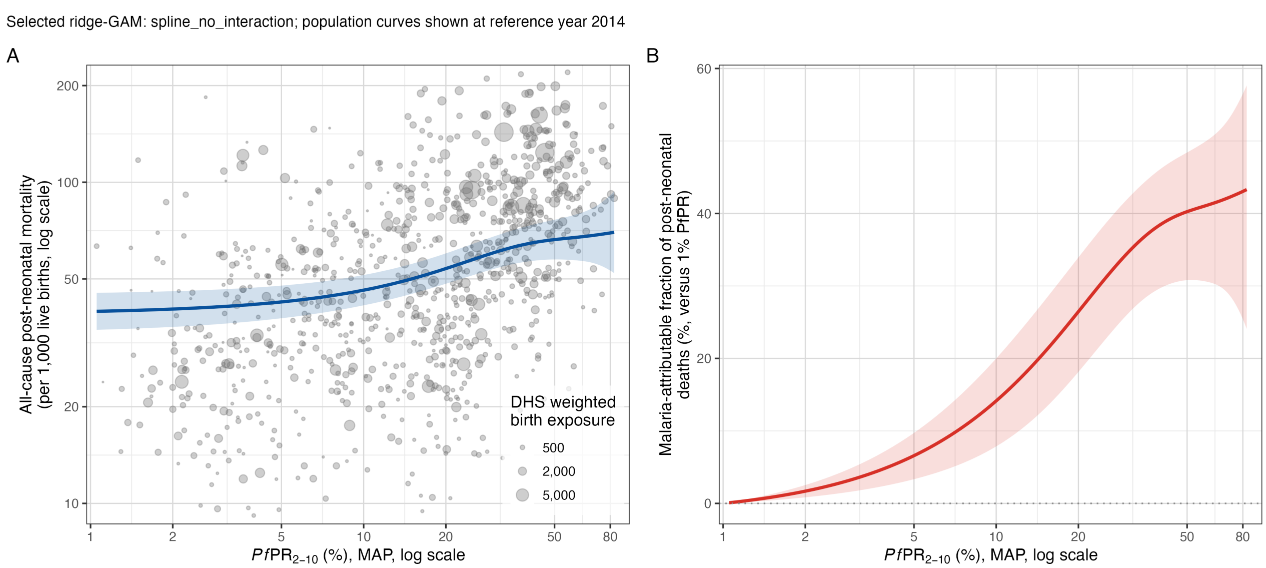

Figure 1: Malaria and post-neonatal mortality. **A:** all-cause post-neonatal mortality versus PfPR2-10, with point size proportional to the survey region's weighted birth exposure and the fitted ridge-GAM curve. **B:** the model-estimated malaria-attributable fraction of post-neonatal deaths as a function of PfPR2-10.

The sections below give the data, methods and supporting detail.

Data and methods

Sample. The unit of analysis is a DHS/MIS survey-region-year (Births Recodes, sub-Saharan Africa, 2000–2024; AIS excluded). Each region receives the population-weighted MAP PfPR2-10 estimate for the survey year; regional all-under-five and neonatal mortality come from birth histories via the DHS synthetic-cohort life-table method, and post-neonatal mortality is their difference. The assembled panel has 936 region-years (106 surveys, 35 countries); the primary sample requires country-mean PfPR2-10 above 1% and regional PfPR2-10 at least 1%, plus positive outcomes/exposure, giving **921 region-years (105 surveys, 34 countries)**. The full data flow is shown in the analysis-flow diagram.

Covariates. Twenty-five candidate covariates are screened before imputation; those with at most 5% missingness are retained and singly imputed (country median, overall-median fallback). The **17 retained covariates** enter a single standardised, ridge-penalised block (proportions logit-transformed):

Level	Covariates
Survey-region (DHS)	urban residence, DTP3, measles, facility delivery, maternal education, exclusive breastfeeding, short birth interval, maternal age at first birth, improved water, improved sanitation, household electricity
National, exact year	WUENIC Hib3, PCV-completion, rotavirus-completion; and child (0–14) HIV prevalence (log)
National, nearest year	log GDP per capita, log health expenditure per capita

Eight candidates are excluded for excess missingness (the three direct DHS vaccine completion recodes, household wealth, the three anthropometry measures, and political stability). National DTP3, electricity access and health-expenditure-share are dropped as redundant with their regional or per-capita counterparts.

Child HIV prevalence is derived from the UNAIDS 2025 estimates workbook (number of children 0–14 living with HIV) divided by the World Bank 0–14 population (`SP.POP.0014.TO`), joined by ISO3 and exact survey year, and entered on the log scale because it spans a very wide range across countries (country means from about 0.02% in Madagascar to about 3.8% in Eswatini). Nigeria and Comoros publish only a 15–19 series (4% of the sample) and are imputed with a provenance flag. Source choices are in `R_dhs/COVARIATE_SOURCES.md`.

Model. Approximate death counts (rate × birth exposure) are modelled with a negative-binomial GAM: `offset(log(exposure))`, the ridge covariate block, country random intercepts and country-specific random linear PfPR slopes. Five PfPR/time specifications are compared by ML/AIC on the same 921 observations, and the selected form is refitted by REML for post-neonatal, all-under-five and neonatal mortality.

Association and model selection

From the ridge-linear summary models, each 10 percentage-point increase in PfPR2-10 is associated with:

Outcome	Change in mortality	95% CI	p-value
Post-neonatal	+8.65%	+5.69% to +11.68%	<0.001
All under-five	+6.37%	+4.01% to +8.79%	<0.001
Neonatal (negative control)	+0.99%	−1.18% to +3.21%	0.37

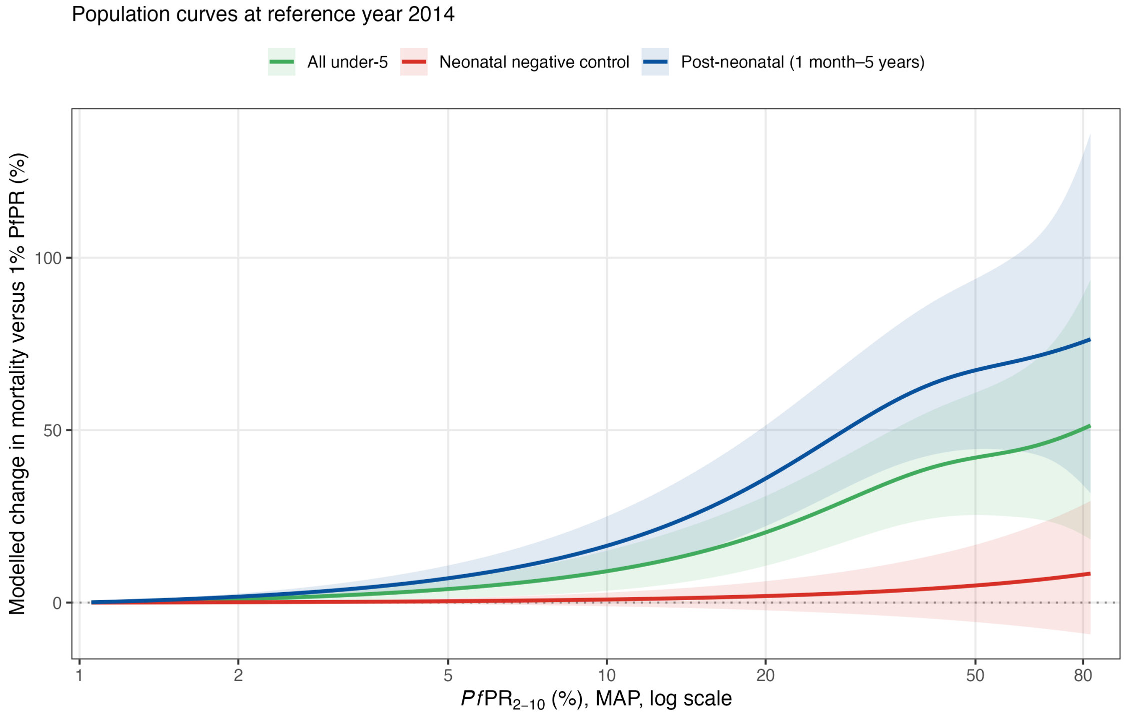

Figure 2: Modelled change in mortality versus 1% PfPR for the three outcomes. The post-neonatal and all-under-five associations rise with prevalence; the neonatal negative control stays flat and its interval spans zero.

Five specifications are compared by AIC on the 921-observation post-neonatal sample:

Model	Specification	AIC	ΔAIC
Spline, no interaction	`s(PfPR) + s(year)`	7100.01	0.00
Spline, `ti` interaction	`s(PfPR) + s(year) + ti(PfPR, year)`	7104.92	4.91
Linear, interaction	`PfPR + PfPR × year + s(year)`	7109.24	9.23
Linear, no interaction	`PfPR + s(year)`	7110.21	10.19
Full tensor surface	`te(PfPR, year)`	7111.23	11.21

An independent AIC comparison for neonatal mortality selects an effectively linear, flat prevalence term; in the prespecified REML refit the neonatal PfPR smooth is linear (edf 1.00) and null (`p=0.37`), consistent with the negative-control table above.

The selected time-stable spline gives the following malaria-attributable fractions of post-neonatal mortality (country random effects set to zero; reference year 2014):

PfPR2-10	Attributable fraction	95% interval
10%	14.1%	7.9%–20.0%
30%	34.4%	26.0%–41.9%
50%	40.2%	30.8%–48.4%

Robustness

Robustness is the central message: the association barely moves across covariate blocks, samples, model structures and likelihood.

- **Covariate adjustment.** Child HIV prevalence is itself a strong predictor of post-neonatal mortality (+15.0% per standard deviation, `p<0.001`), yet adding it leaves the malaria estimate essentially unchanged and only weakens the PfPR-by-time interaction (`p` 0.08→0.16). The regional block alone and the full regional-plus-national block give a similar attributable fraction (about 34–36% at 30% PfPR).

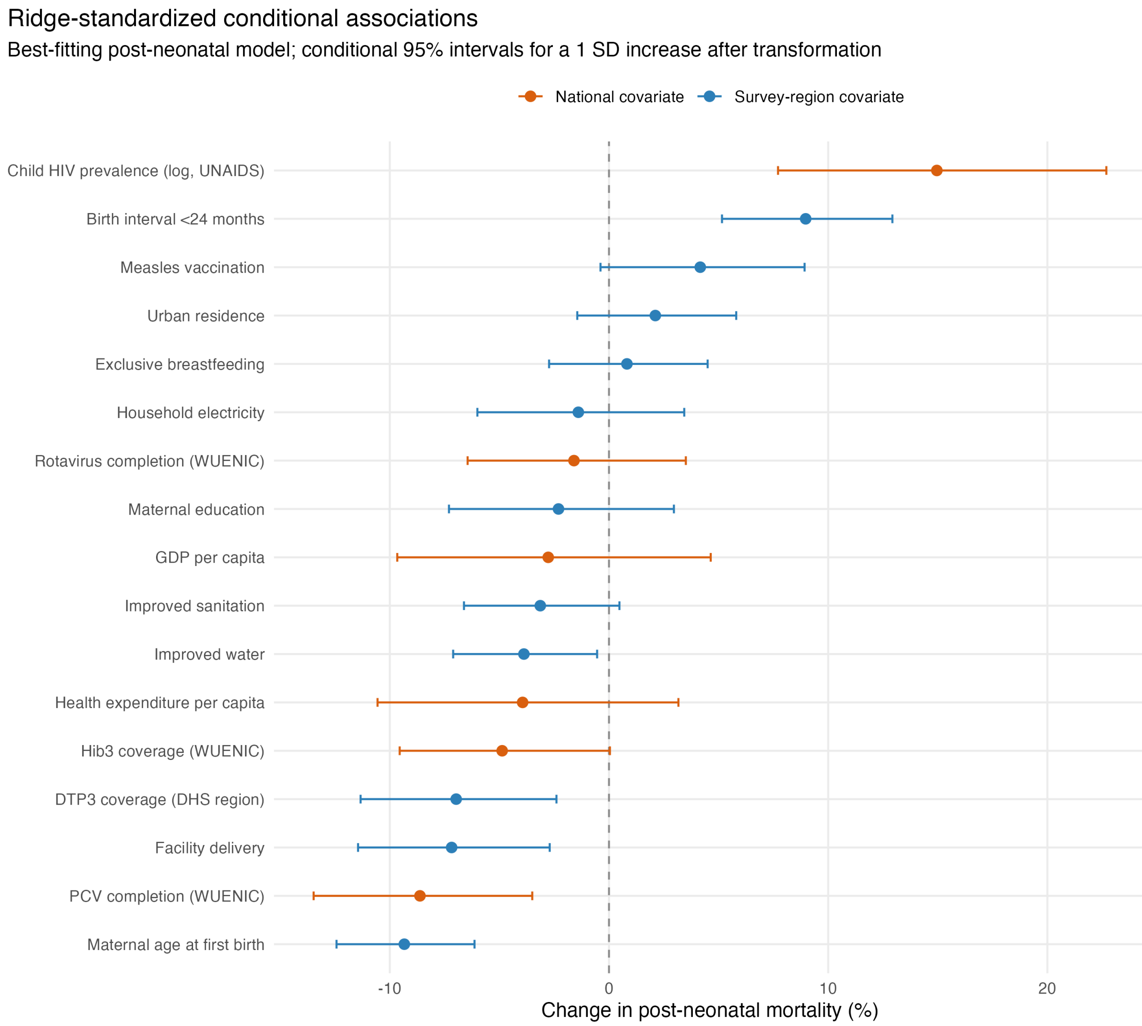

Figure 3: Ridge-standardized conditional associations for the 17 covariates in the best-fitting post-neonatal model (percentage change per 1 SD after transformation; ridge-shrunk, not independent causal effects).

- **Samples, structure and likelihood.** The attributable fraction at 30% PfPR stays near 34% across alternative samples (including sub-1% regions, the 5–40% range, and complete cases) and across model-structure choices; it rises materially only in a national-covariates-only model that fits far worse. A log-Gaussian rate model is less precise but still positive (+5.7% per 10 PfPR points; `p=0.14`).

External validation: bednet trials

An independent triangulation against eight insecticide-treated-net/curtain cluster-RCTs — which measured both parasite prevalence and child mortality — recovers essentially the same slope as the observational model: a weighted meta-regression gives about an 8.6% mortality reduction per 10-point drop in age-standardised PfPR2-10, matching the model's ridge-linear slope (though imprecise — `p=0.12`, eight heterogeneous trials). Predicted and observed trial effects broadly track each other.

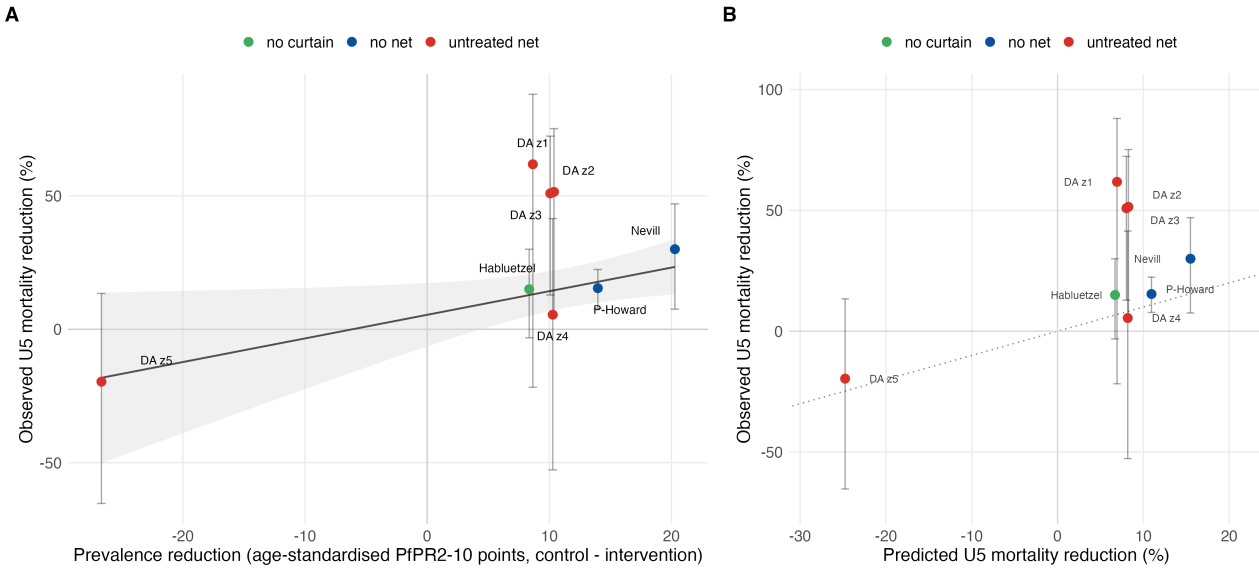

Figure 4: RCT triangulation. **A:** observed under-five mortality reduction versus the age-standardised prevalence reduction across trial arms (weighted fit). **B:** mortality reductions predicted from the model's prevalence slope versus observed (dotted line = 1:1).

National burden

Latest-year comparison

The comparison is not perfectly contemporaneous: the model surface is evaluated at 2025 using the latest available 2024 MAP PfPR, IGME all-cause mortality and live births for 43 matched sub-Saharan African countries; IHME is 2025 under-five malaria deaths; WHO WMR 2025 reports all-age country deaths for 2024, so the under-five comparator applies WHO's ~75% age-share statement.

Source	Deaths
Selected time-stable model	**482,000**
IHME/GBD SSA under-five, 2025	470,000
WHO African-region all-age, 2024	579,000
WHO African-region under-five proxy, 2024	434,000

Temporal trend

The additive spline is time-stable, but the time-varying alternatives — although not selected — remain the main sensitivity for the historical trend. Applied to national PfPR and all-cause mortality series, estimated post-neonatal malaria deaths from 2000 to 2024 fall by:

Model	2000	2024	Change
No interaction (selected)	1,012,000	482,000	−52%
Spline `ti` interaction	878,000	527,000	−40%
Full `te` surface	802,000	551,000	−31%

Under the selected model the attributable fraction is time-stable; the time-varying alternatives instead let it rise gradually over the period. Adjusting for child HIV prevalence — itself a large time-varying driver of child mortality through ART rollout — brings the time-varying declines closer to the time-stable one than in the unadjusted analysis.

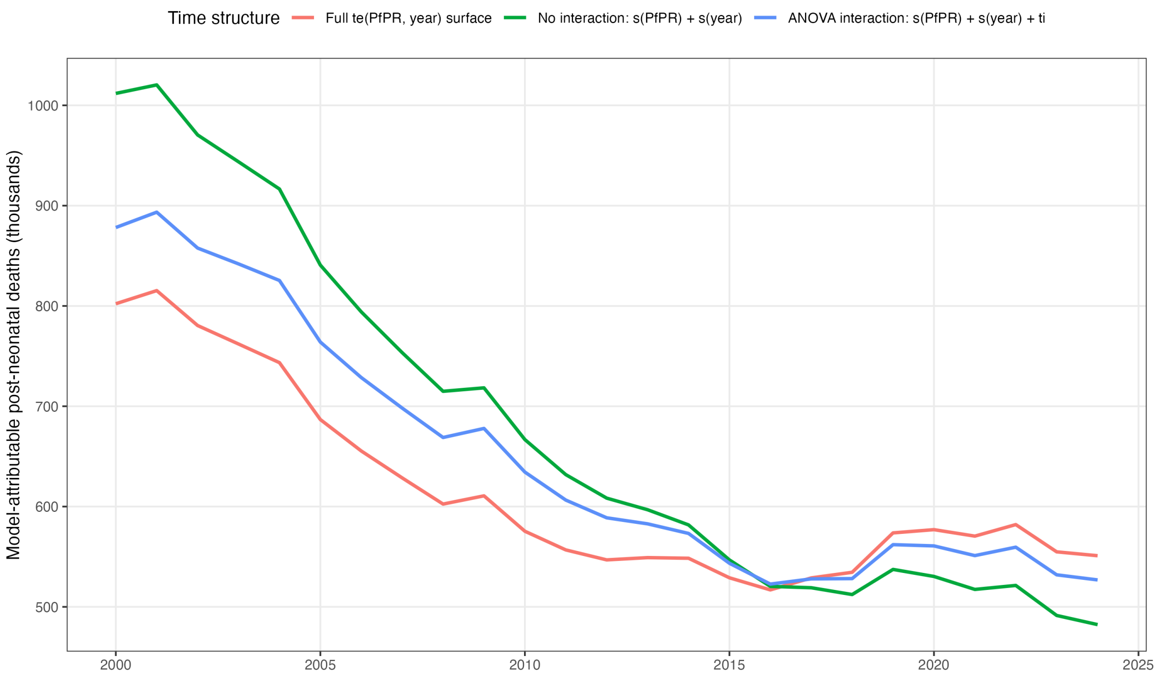

Figure 5: National malaria-attributable post-neonatal deaths over time under the three time structures.

Country differences from IHME

The aggregate agreement with IHME masks large country-level differences (AIC-selected time-stable spline; “very different” = at least two-fold and 1,000 deaths):

Country	Model	IHME 2025	Model/IHME
DR Congo	110,000	68,300	1.61
Nigeria	178,000	150,500	1.18
Chad	12,300	5,500	2.23
Sudan	3,900	1,400	2.89
Uganda	13,400	28,500	0.47
Burundi	3,900	15,700	0.25
Tanzania	5,100	10,700	0.47
Ethiopia	4,000	9,300	0.42
Ghana	3,500	7,300	0.49
Rwanda	300	2,500	0.11

DR Congo and Nigeria account for most of the model’s upward difference from IHME, offset by lower estimates elsewhere; this is why the aggregate totals are close despite the country-level spread.

Interpretation and limitations

The data provide fairly robust evidence of a positive cross-sectional association between malaria prevalence and post-neonatal mortality that survives adjustment for national vaccine rollout and child HIV prevalence, the neonatal negative control, alternative samples, and model-structure choices. The main remaining limitations are:

- the aggregate burden is not a validation against a common estimand — the model includes indirect malaria-associated deaths, whereas IHME and WHO assign deaths specifically to malaria, and national extrapolation uses population-average effects that ignore country random effects;

- 2025 exposure and all-cause mortality inputs are unavailable, and the WHO under-five comparison applies a regional 75% age-share to every country;
- child HIV prevalence is derived from UNAIDS 0–14 counts and a World Bank denominator rather than a published rate, and Nigeria and Comoros (about 4% of the sample, including the largest-burden country) lack an under-15 series and are imputed;
- the log-Gaussian likelihood sensitivity is positive but imprecise, so the exact effect size and the temporal trend remain uncertain.

Analysis flow

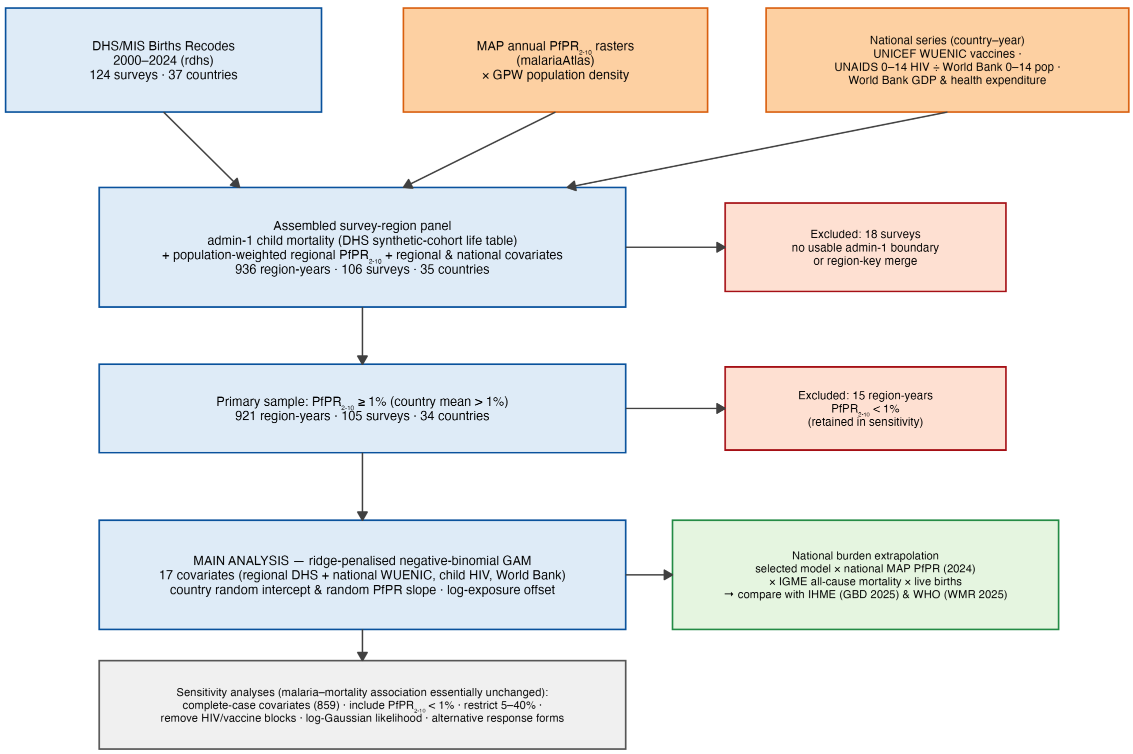

Figure 6: Data and analysis flow. External data (MAP PfPR2-10 rasters, UNICEF WUENIC vaccine coverage, UNAIDS/World Bank child HIV prevalence, and World Bank economic series) feed the assembled survey-region panel, which is filtered to the primary sample, analysed with the ridge-penalised model, and extrapolated to a national burden.